

al. [14], enhance fuel efficiency but impose high computational costs due to system nonlinearities. An alternative approach, the Equivalent Consumption Minimization Strategy (ECMS) [25], reduces computational complexity while delivering near-optimal fuel savings by relying on instantaneous slope data. Both predictive and non-predictive strategies are viable for connected automated vehicles (CAVs), yet achieving a balance between fuel efficiency and computational feasibility remains a key challenge for practical deployment.

## 1.2 Outline of the paper

In Section 2 we introduce a convex relaxation of the problem. In Section 3 we briefly sketch the exactness result for this relaxation (under suitable assumptions) when we drop the boundary condition  $w_n = w_{\text{fin}}$  at step  $n$ , already proved in our paper [2]. Moreover, we show that the relaxation is not exact any more as soon as we include also the boundary condition. In Section 4 we introduce a feasibility-based bound tightening technique, while in Section 5 we prove that the addition of the lower limit for the variables obtained through the bound tightening procedure makes the convex relaxation exact. Next, in Section 6 we introduce further feasibility-based bound tightening techniques. In Section 7 we prove that the feasible region of our problem is a lattice and from that we derive a necessary and sufficient condition to establish its non-emptiness, also discussing a procedure to verify such condition, based on the iterated application of the bound tightening techniques discussed in Sections 4 and 6. Finally, in Section 8 we draw some conclusions.

## 2 Convex relaxation of the problem

The constraints of our problem are the following:

$$\frac{M}{P_{\max}\sqrt{w_i}} \geq \frac{1}{h}(w_{i+1} - w_i) + \gamma w_i + g(\sin \alpha_i + c) \quad i \in \{1, \dots, n-1\} \quad (4)$$

$$\frac{1}{h}(w_{i+1} - w_i) + \gamma w_i + g(\sin \alpha_i + c) \leq g\mu \quad i \in \{1, \dots, n-1\} \quad (5)$$

$$\frac{1}{h}(w_{i+1} - w_i) + \gamma w_i + g(\sin \alpha_i + c) \geq -g\mu \quad i \in \{1, \dots, n-1\} \quad (6)$$

$$w_i \leq w_i^{\max} \quad i \in \{1, \dots, n\} \quad (7)$$

$$w_i \geq 0 \quad i \in \{1, \dots, n\} \quad (8)$$

$$w_1 = w_{\text{init}}, \quad w_n = w_{\text{fin}}. \quad (9)$$

We rewrite constraints (7)–(9) as follows:

$$w_i \leq w_i^{\max} \quad i \in \{1, \dots, n\} \quad (10)$$

$$w_i \geq w_i^{\min} \quad i \in \{1, \dots, n\}, \quad (11)$$

where  $w_i^{\min} = 0$ , for  $i \in \{2, \dots, n-1\}$ , while  $w_1^{\min} = w_1^{\max} = w_{\text{init}}$  and  $w_n^{\min} = w_n^{\max} = w_{\text{fin}}$  (note that the two boundary conditions (9) are split into the constraints  $w_{\text{init}} \leq w_1 \leq w_{\text{init}}$  and  $w_{\text{fin}} \leq w_n \leq w_{\text{fin}}$ ). Moreover, after introducing the variables  $t_i, f_i$ ,  $i = 1, \dots, n-1$ , we can replace the maximum power constraints (4) with the following constraints:

$$t_i = \frac{1}{\sqrt{w_i}} \quad i \in \{1, \dots, n-1\} \quad (12)$$

$$t_i \geq \frac{Mf_i}{P_{\max}} \quad i \in \{1, \dots, n-1\} \quad (13)$$

$$f_i = \frac{1}{h}(w_{i+1} - w_i) + \gamma w_i + g(\sin \alpha_i + c) \quad i \in \{1, \dots, n-1\}. \quad (14)$$